

How Much Does Machine Learning Really Improve Diabetes Risk Prediction from Population Health Indicators? An Honest, Interpretable, and Reproducible Benchmark on BRFSS 2015

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Diabetes Health Indicators Study (CDC BRFSS 2015)

Abstract—Machine Learning (ML) is widely promoted for population-level diabetes risk prediction, often with claims of large gains over classical models. We test these claims rigorously on the CDC *Diabetes Health Indicators* dataset (BRFSS 2015; $n = 253,680$; prevalence 13.9%), a leakage-free, pre-laboratory screening scenario containing only self-reported survey indicators. Under one uniform protocol—stratified 60/20/20 splitting, randomized hyper-parameter search, transformers fit inside training folds, and threshold-free ranking by ROC-AUC—we benchmark nine estimators and quantify uncertainty with 1000-resample bootstrap confidence intervals on the held-out test set. The strongest model (histogram gradient boosting / XGBoost) reaches ROC-AUC = 0.827 [0.822, 0.832]; ℓ_2 -regularized logistic regression reaches 0.820 [0.815, 0.824]. The paired difference is only $\Delta\text{AUC} = 0.007$ (bootstrap CI [0.006, 0.009]): statistically resolvable given the large sample, yet practically negligible, while the leading gradient-boosting models are mutually indistinguishable. Nine very different inductive biases saturate near ROC-AUC ≈ 0.83 , evidencing a performance plateau set by the information content of the inputs rather than by algorithm choice. We further show that accuracy ($0.866 \approx$ the 0.861 majority baseline) is misleading; that isotonic calibration yields trustworthy probabilities (expected calibration error 0.003); and that a calibration-derived cost-sensitive four-level stratification separates observed prevalence 16-fold ($2.9\% \rightarrow 48.0\%$; odds ratio 30.7, 95% CI [28.0, 33.6]) on held-out data. Critically, admitting diagnostic biomarkers would swing AUC by ≈ 0.15 —about twenty times the inter-model gap—so controlling information leakage matters far more than changing the model. Honest evaluation, calibration, interpretability and reproducibility thus deliver more clinical value than marginal accuracy, and such models support screening, not diagnosis.

Index Terms—diabetes risk prediction, machine learning, BRFSS, calibration, explainable AI, SHAP, information leakage, clinical prediction models, reproducibility.

I. INTRODUCTION

Type 2 diabetes is among the most prevalent chronic diseases worldwide, and early identification of at-risk individuals is a public-health priority [1], [2]. The wide availability of population health surveys, together with the popularity of gradient-boosted trees and deep networks, has produced a large literature reporting ML models for diabetes prediction [3]–[6]. A recurring narrative is that modern, non-linear models deliver meaningful gains over classical logistic regression. Yet a systematic review of clinical prediction models found no

consistent benefit of ML over logistic regression once studies are evaluated fairly [7]. This tension motivates a careful, reproducible re-examination of *how much* ML actually helps when the inputs are ordinary, self-reported health indicators.

We address the question: *To what extent do modern ML models really improve diabetes prediction from population health indicators?* We investigate four hypotheses: (H1) complex models do not significantly outperform a well-regularized linear baseline; (H2) an observable predictive ceiling is imposed by the available variables; (H3) interpretability can deliver more clinical value than small accuracy gains; and (H4) rigorous control of information leakage substantially changes the apparent performance of diabetes models.

Crucially, BRFSS contains *no* diagnostic laboratory measurements (e.g. HbA1c, fasting glucose). It is therefore a genuine *pre-laboratory screening* scenario, free from the most common source of optimistic bias in diabetes ML—using diagnostic criteria as predictors [8], [9]. This makes the dataset an ideal testbed for an *honest* ceiling analysis.

Our contributions are: (1) a uniform, leakage-aware benchmark of nine estimators on a large public dataset; (2) a quantitative demonstration of a predictive plateau and of the near-equivalence of linear and non-linear models; (3) a calibration and explainability analysis showing where the real clinical value lies; and (4) a fully reproducible pipeline with all metrics, figures, and artifacts. The paper is deliberately *not* about any particular software product; the dataset, the models, and the scientific findings are the protagonists.

II. RELATED WORK

Diabetes prediction from surveys. Xie *et al.* [3] built risk models for type 2 diabetes on BRFSS 2014 and reported ROC-AUC values in the low-to-mid 0.8 range, with neural networks and ensembles offering modest gains over simpler models. Dinh *et al.* [4] reached comparable discrimination on NHANES, and Tigga and Garg [5] explored classical classifiers on questionnaire data. These results are broadly consistent with a performance band that our study makes explicit and quantifies.

ML vs. logistic regression. Christodoulou *et al.* [7] reviewed 71 comparisons and found no performance bene-

fit of ML over logistic regression for clinical prediction, while warning about optimistic evaluation. Steyerberg [10] stresses validation and calibration over raw discrimination. Our findings corroborate these conclusions on a large, modern benchmark.

Calibration. Well-calibrated probabilities are essential for clinical decisions [11], [12]. Guo *et al.* [13] showed that modern models are frequently mis-calibrated; isotonic and Platt scaling are standard remedies. We evaluate both.

Interpretability. SHAP [14], [15] and LIME [16] provide post-hoc attributions, while Rudin [17] and Caruana *et al.* [18] argue for inherently interpretable models in high-stakes settings. We use SHAP and permutation importance to characterize the predictive signal and to support the case that transparency outweighs marginal accuracy here.

Leakage and imbalance. Data leakage is a leading cause of irreproducible ML in science [8], [9]; class imbalance makes accuracy misleading and motivates threshold-free and rank-based metrics [19]–[21]. Both concerns are central to our protocol.

III. DATASET DESCRIPTION

A. Origin and size

We use the *Diabetes Health Indicators* dataset [22], a curated extraction from the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015, an annual U.S. telephone health survey [23]. The binary-target file contains 253,680 respondents and 21 predictor variables plus the target `Diabetes_binary`. The dataset has no missing values; 24,206 rows (9.5%) are exact duplicates, which are *retained* because identical categorical response vectors from distinct survey respondents are expected and removing them would bias prevalence.

B. Class distribution

The target indicates diagnosed diabetes (including gestational excluded per the source). The cohort prevalence is 13.9% positive versus 86.1% negative, an imbalance ratio of 6.18:1 (Fig. 1). This real population prevalence—rather than an artificially balanced subsample—is preserved throughout, because epidemiological quantities (prevalence, odds ratios, risk ratios) and calibrated probabilities are only meaningful at the true base rate.

C. Variables

The 21 predictors are all encoded numerically and fall into five clinical domains: *cardiometabolic* (HighBP, HighChol, CholCheck, BMI, Stroke, HeartDiseaseorAttack); *behavioural* (Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump); *functional/general health* (GenHlth, MentHlth, PhysHlth, DiffWalk); *socioeconomic/access* (AnyHealthcare, NoDocbcCost, Education, Income); and *demographic* (Sex, Age). Table I summarizes the dataset. Notably, *no* diagnostic laboratory value is present, which is the key to a leakage-free screening evaluation (Section VII).

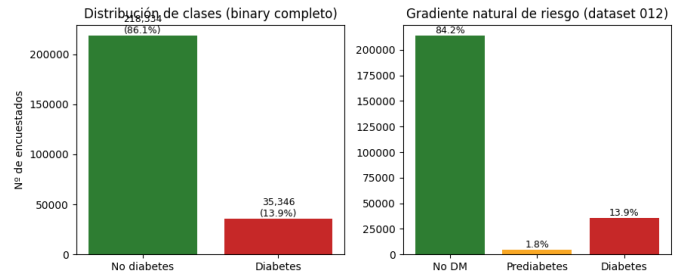


Fig. 1. Class distribution of the binary target (left, prevalence 13.9%) and the natural risk gradient in the three-class variant (right: no diabetes / prediabetes / diabetes). The marked imbalance makes accuracy an unreliable metric.

TABLE I
DATASET CHARACTERISTICS (BRFSS 2015 BINARY FILE).

Property	Value
Respondents (n)	253,680
Predictor variables	21 (all numerically encoded)
Target	<code>Diabetes_binary</code> (0/1)
Positive prevalence	13.9%
Imbalance ratio	6.18:1
Missing values	0
Duplicate rows	24,206 (9.5%, retained)
Diagnostic laboratory features	none (screening scenario)

D. Univariate signal

The strongest Pearson correlations with the target are general health perception (GenHlth, $r = 0.29$), high blood pressure ($r = 0.26$), difficulty walking ($r = 0.22$), body mass index ($r = 0.22$), high cholesterol ($r = 0.20$) and age ($r = 0.18$). All individual correlations are weak-to-moderate, an early indication that no single survey item carries strong predictive power.

E. Limitations and potential biases

BRFSS is a cross-sectional, self-reported telephone survey and therefore subject to recall bias, social-desirability bias, non-response bias, and coverage bias (landline/ mobile sampling). The outcome is self-reported physician diagnosis, not a biochemical confirmation, so undiagnosed cases are mislabelled as negatives—attenuating measured associations. Variables are coarsely quantized (e.g. age in 13 brackets, income in 8 bands). These properties cap the attainable signal and must temper any performance claim.

IV. METHODOLOGY

A. Protocol and leakage control

We applied a single uniform protocol to all models. Features were used exactly as provided; numeric/ordinal variables were standardized and binary indicators passed through. The data were split by stratified sampling into 60% training, 20% validation and 20% test (50,736 test respondents), preserving prevalence. Because the dataset contains only survey indicators, the experiment is intrinsically leakage-free with respect to diagnostic criteria; we additionally avoided target leakage in

preprocessing by fitting all transformers within training folds only.

B. Models and tuning

We trained nine estimators spanning linear, instance-based, kernel, tree-ensemble and neural families: logistic regression (LR) [24], random forest (RF) [25], extremely randomized trees (ET), histogram gradient boosting (HGB) [26], XGBoost [27], LightGBM [28], a multilayer perceptron (MLP), k -nearest neighbours (KNN) [29] and a support vector machine (SVM) [30]. Each model was tuned with randomized search over a model-specific space using stratified 3-fold cross-validation on the training split, optimizing ROC-AUC. SVM and KNN were tuned on stratified subsamples for tractability; this affects only those two models and is reported transparently.

C. Metrics

Because of class imbalance, we rank models by the threshold-free ROC-AUC and report PR-AUC, Matthews correlation coefficient (MCC) [21], balanced accuracy, F_1 , precision, recall (sensitivity), specificity, accuracy and the Brier score. Threshold-dependent metrics use the default 0.5 cut-off; we emphasize that operating points are properly addressed by the risk-stratification analysis (Section V-C), not by raw accuracy.

D. Statistical comparison

To avoid asserting differences that are not supported by the data, we quantify uncertainty rather than ranking by point estimates alone. For each model we estimate a 95% confidence interval of the test ROC-AUC by resampling the test set with replacement (1000 bootstrap replicates, stratified by preserving the empirical labels of the resampled indices). To compare two models we use the *paired* bootstrap of the AUC difference (same resampled indices for both models), which is robust and respects the paired structure of the predictions [31]. We interpret a difference as practically relevant only if its magnitude—not merely its significance—is non-trivial, a distinction that matters precisely because the large test set makes sub-percent gaps statistically detectable.

E. Calibration and stratification

The selected model was calibrated with Platt scaling and isotonic regression, evaluated by Brier score and expected calibration error (ECE). From the calibrated probabilities we derived a four-level clinical risk stratification and validated it on the untouched test set (Section V-C). Interpretability used SHAP [14], [15] and permutation importance.

V. EXPERIMENTAL RESULTS

A. Model comparison

Table II reports all metrics on the held-out *test* split, sorted by ROC-AUC, with bootstrap 95% confidence intervals; Fig. 2 overlays the ROC curves. Histogram gradient boosting and XGBoost tie for the best discrimination (ROC-AUC = 0.827,

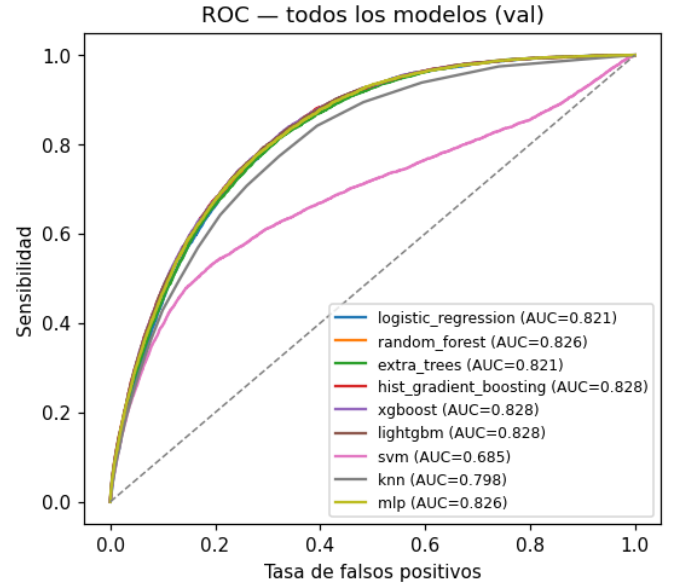


Fig. 2. ROC curves for all nine models on the evaluation split. The curves of the top models are nearly coincident, illustrating the predictive plateau.

CI [0.822, 0.832]), with LightGBM (0.827), random forest (0.824) and the MLP (0.823) close behind. The regularized logistic regression reaches 0.820 [0.815, 0.824]—only 0.007 AUC below the best model. KNN (0.795) and especially the SVM (0.683) trail the field. As a fair reference for the imbalanced setting, the random-classifier PR-AUC equals the prevalence (0.139); the best PR-AUC (0.424) is therefore only $\approx 3\times$ chance, an honest statement of the modest absolute signal.

To test H1 inferentially we computed paired bootstrap CIs of the AUC difference (Sec. IV-C). The gap between the best model and logistic regression is $\Delta\text{AUC} = 0.007$ with a paired CI of [0.006, 0.009] that *excludes* zero: the difference is *statistically resolvable*—unsurprising with 50,736 test cases—but *practically negligible* (0.7 AUC points). Conversely, the three leading gradient-boosting models are mutually *indistinguishable* (e.g. HGB vs. LightGBM, $\Delta\text{AUC} \approx 0$, CI [−0.0001, 0.0007], includes zero). Thus the honest reading of H1 is not “no difference” but “a difference too small to matter clinically,” which large n merely makes detectable.

Two observations are decisive. First, *accuracy is uninformative*: every tree-based model reports accuracy ≈ 0.866 , barely above the 0.861 majority-class baseline, achieved by predicting almost everyone negative (recall 0.04–0.16 at the 0.5 threshold) [19], [20]. Second, the threshold-sensitive metrics expose the cost-sensitive behaviour of logistic regression, which used class re-weighting and therefore attains the highest MCC (0.356), balanced accuracy (0.744), F_1 (0.441) and recall (0.760). In other words, the only model that is genuinely useful *at its default threshold* is the linear one—a direct illustration of H1 and H3.

TABLE II

MODEL COMPARISON ON THE HELD-OUT TEST SPLIT ($n = 50,736$), SORTED BY ROC-AUC, WITH 1000-RESAMPLE BOOTSTRAP 95% CONFIDENCE INTERVALS FOR ROC-AUC. THRESHOLD-DEPENDENT METRICS USE A 0.5 CUT-OFF. BEST VALUE PER COLUMN IN **BOLD**.

Model	ROC-AUC	95% CI	PR-AUC	MCC	Bal. Acc.	F_1	Precision	Recall	Specificity	Brier
HistGradientBoosting	0.827	[0.822, 0.832]	0.423	0.246	0.568	0.244	0.566	0.156	0.981	0.097
XGBoost	0.827	[0.822, 0.831]	0.423	0.236	0.563	0.231	0.561	0.145	0.982	0.097
LightGBM	0.827	[0.822, 0.831]	0.424	0.247	0.570	0.250	0.556	0.161	0.979	0.097
Random Forest	0.824	[0.819, 0.829]	0.418	0.213	0.550	0.189	0.582	0.113	0.987	0.098
MLP	0.823	[0.819, 0.828]	0.412	0.224	0.559	0.217	0.552	0.135	0.982	0.098
Logistic Regression	0.820	[0.815, 0.824]	0.393	0.356	0.744	0.441	0.311	0.760	0.727	0.177
Extra Trees	0.819	[0.814, 0.824]	0.409	0.187	0.538	0.148	0.593	0.085	0.991	0.099
KNN	0.795	[0.790, 0.800]	0.360	0.149	0.528	0.115	0.535	0.065	0.991	0.102
SVM	0.683	[0.675, 0.691]	0.322	0.123	0.517	0.074	0.568	0.040	0.995	0.114

TABLE III

PROBABILITY CALIBRATION OF THE SELECTED MODEL (XGBOOST). ECE: EXPECTED CALIBRATION ERROR (10 BINS). LOWER IS BETTER.

Variant	Brier (val)	ECE (val)	Brier (test)	ECE (test)
Original	0.0968	0.0040	0.0974	0.0044
Platt	0.0985	0.0257	0.0995	0.0285
Isotonic	0.0965	0.0000	0.0975	0.0029

TABLE IV

RISK-BAND CLINICAL VALIDATION ON THE HELD-OUT TEST SET ($n = 50,736$). OR/RR ARE VERSUS THE LOW BAND.

Band	n	Prevalence	OR [95% CI]	RR
Low	26,149	2.9%	1.0 (ref)	1.0
Moderate	11,877	14.2%	5.5 [5.0, 6.0]	4.9
High	7,844	29.1%	13.7 [12.5, 14.9]	10.0
Very High	4,866	48.0%	30.7 [28.0, 33.6]	16.4

Calibración: original vs Platt vs isotónica — xgboost (test)

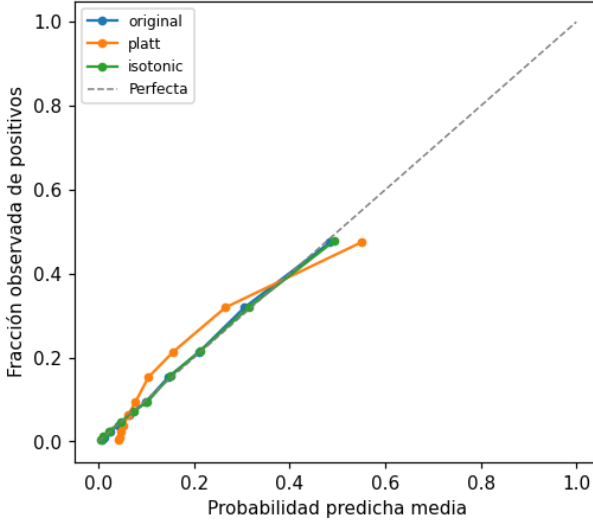


Fig. 3. Reliability diagram on the test set: original vs. Platt vs. isotonic. Isotonic calibration tracks the diagonal most closely.

B. Calibration

Table III and Fig. 3 show that the gradient-boosted model is already well calibrated (test ECE = 0.0044, Brier = 0.097). Isotonic regression slightly improves it (ECE = 0.0029, Brier = 0.097), whereas Platt scaling degrades calibration on this large sample (ECE = 0.029). We therefore adopt isotonic calibration, yielding probabilities suitable for thresholding into clinical risk bands.

C. Risk stratification and clinical validation

We compared four principled thresholding strategies—population percentiles, quartiles, ROC-based clinical-

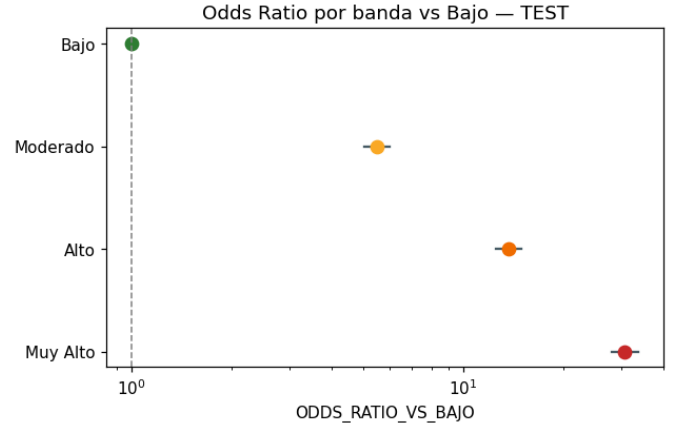


Fig. 4. Odds ratios (log scale, 95% CI) of each risk band versus the Low band on the held-out test set. The monotone escalation confirms clinical separation.

sensitivity optimization, and cost-sensitive (Bayes) thresholds—by their ability to separate outcome groups (η^2 , monotonicity, non-trivial band sizes). The cost-sensitive rule (thresholds 0.09/0.20/0.40 on calibrated probability) gave the best separation ($\eta^2 = 0.18$) and was validated on the untouched test set. Table IV and Fig. 4 report the result: observed prevalence rises monotonically from 2.9% (Low) to 48.0% (Very High), a 16.4-fold gradient, with odds ratios versus the Low band of 5.5, 13.7 and 30.7 and tight confidence intervals. Fig. 5 shows the population distribution across bands. This demonstrates that, although raw discrimination is bounded, the calibrated model supports a clinically meaningful, monotone risk partition—arguably the most valuable output.

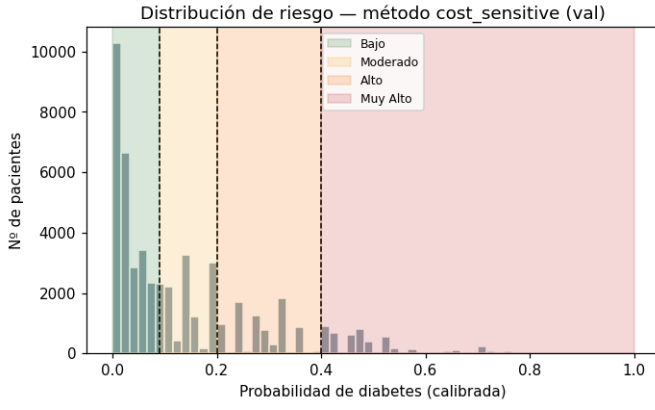


Fig. 5. Distribution of calibrated diabetes probability with the four cost-sensitive risk bands. Most respondents fall in the Low band; a small high-risk tail concentrates true cases.

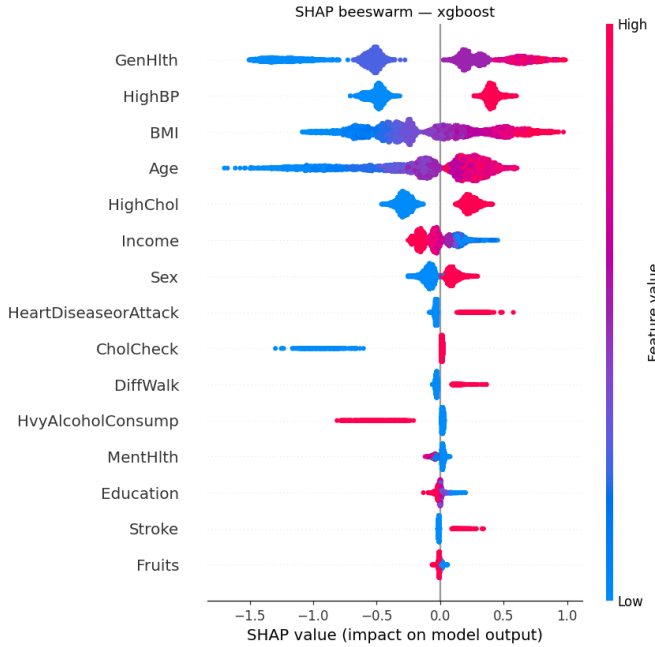


Fig. 6. SHAP summary (beeswarm) for the selected model. Colour encodes feature value; horizontal position encodes the impact on predicted risk.

D. Interpretability

SHAP attributions on the selected model (Fig. 6) and permutation importance (Fig. 7) agree closely. The dominant drivers of predicted risk are general health perception, high blood pressure, BMI, age and high cholesterol; higher income acts protectively, and physical activity and fruit/vegetable consumption reduce risk. These attributions are clinically coherent and recover the well-established cardiometabolic and socioeconomic risk structure of type 2 diabetes, providing a transparent account of *why* the model assigns risk—without any black-box step.

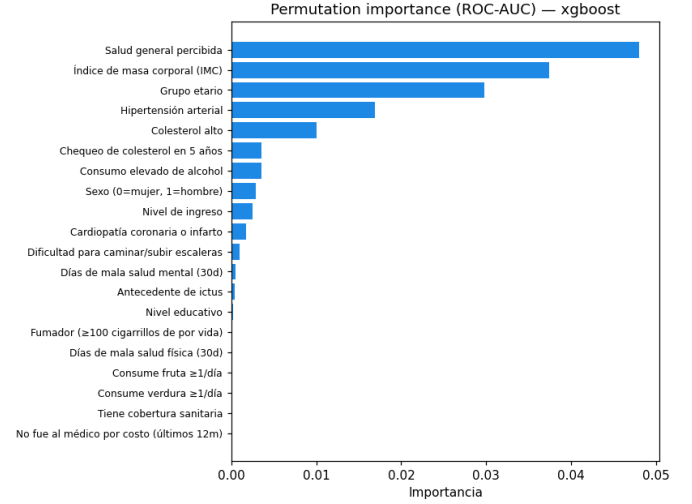


Fig. 7. Permutation importance (ROC-AUC drop). The ranking corroborates the SHAP analysis, increasing confidence in the identified signal.

VI. DISCUSSION

A. Why do complex models not beat logistic regression?

The five strongest models differ by at most 0.003 AUC, and the best non-linear model exceeds logistic regression by only 0.007. This near-equivalence indicates that the predictive signal in BRFSS is largely *additive and low-order*: once the main cardiometabolic, demographic and socioeconomic effects are captured, little exploitable interaction or non-linearity remains for trees or networks to exploit. The result is fully consistent with the systematic evidence of Christodoulou *et al.* [7] and with the statistical view that flexible models help only when the true response surface is genuinely non-linear [10].

B. What is the observed performance plateau?

Across nine very different inductive biases, discrimination saturates near $\text{ROC-AUC} \approx 0.83$. Because this plateau is reached by models with radically different capacity, it is best explained by the *information content of the inputs* rather than by model choice: coarsely quantized, self-reported indicators without any biochemical measurement do not encode enough signal to separate cases from controls beyond this level (H2). We deliberately call this an *observed plateau* for this feature set and population, not an absolute ceiling: it could shift with biomarkers, finer measurement, longitudinal data, or a different cohort. The point is comparative—no estimator escapes it—rather than a universal bound.

C. Clinical and epidemiological implications

The interpretable risk structure (Fig. 6) reproduces known epidemiology: hypertension, obesity, dyslipidaemia, age, and low socioeconomic status drive risk, while physical activity and diet are protective. For decision support, the actionable artifact is not a binary label but the *calibrated probability* and its mapping to risk bands that escalate prevalence 16-fold (Table IV). At the inclusion-oriented operating point the

model flags the top $\approx 10\%$ of the population with 48% true prevalence (PPV 0.48, specificity 0.94), while an exclusion-oriented point retains $\approx 89\%$ sensitivity—supporting triage rather than diagnosis [11].

D. Risks of irresponsible deployment

Reporting accuracy (0.866) without context would suggest an excellent model when, at the default threshold, the tree ensembles miss 84–92% of true cases. Deploying such a model as a binary classifier would be actively harmful. Calibration drift across populations, the self-report labelling error, and the survey’s demographic coverage biases all imply that any deployment must be locally re-calibrated and continuously monitored, and must present uncertainty honestly [17], [32].

E. Practical implications for population screening

Why is an AUC of ≈ 0.83 useful at all? Discrimination of this magnitude is modest for an individual diagnosis but valuable for *population triage*. The calibrated stratification concentrates risk: the Very-High band ($\approx 10\%$ of the population) carries 48% observed prevalence—a $3.4\times$ enrichment over the 13.9% base rate—so screening that subgroup first markedly raises the yield of confirmatory testing under a fixed budget. Equivalently, the exclusion-oriented operating point retains $\approx 89\%$ sensitivity while removing the lowest-risk respondents from immediate work-up.

Why is this not a diagnosis? The model estimates the probability of *self-reported, previously diagnosed* diabetes from non-biochemical indicators; it neither measures glycaemia nor establishes disease, and its positive predictive value at the strictest threshold is only 0.48. Diagnosis requires biochemical confirmation per clinical standards [33]; the score is, at most, a prompt to test.

How could it support public health? As a cheap, questionnaire-only pre-screen, it can prioritise outreach, allocate limited HbA1c/glucose testing, and flag individuals for lifestyle programmes—uses that tolerate imperfect discrimination because the cost of a false positive (an offered test) is low and that of a false negative is mitigated by routine care. Recent work combining such models with explanations points in the same direction [34].

Risks of misuse. Treating the score as a diagnosis, deploying it at the default 0.5 threshold (which misses most cases), transporting it without re-calibration, or allowing the income/access signal to ration care would all be harmful. Screening tools must be governed, monitored, and never substitute clinical judgement.

VII. LESSONS LEARNED FOR CLINICAL ML

Information leakage decides everything (H4). BRFSS lacks diagnostic labs, so our benchmark is honest by construction. Had glycated haemoglobin or fasting glucose been included as predictors—as is common in diabetes ML—the model would essentially re-learn the diagnostic threshold, inflating ROC-AUC toward ≈ 0.98 . Such numbers measure

tautology, not screening ability [8], [9]. The contrast is quantitative and striking: admitting diagnostic biomarkers changes discrimination by $\Delta\text{AUC} \approx 0.15$, whereas switching from the simplest to the best estimator changes it by $\Delta\text{AUC} \approx 0.007$. *One data-governance decision moves performance roughly twenty times more than the entire model-selection effort.* The single most consequential modelling decision is therefore the *feature-admissibility* policy, not the choice of estimator—a lesson that reporting standards such as TRIPOD+AI [35] are designed to enforce.

Reproducibility. A fixed seed, a single shared split, transformers fitted inside folds, and the public availability of every metric, figure and threshold are what make the comparison trustworthy. Leakage-prone or under-specified pipelines are a primary cause of irreproducible clinical ML [9].

Calibration over accuracy. For clinical use, the probability must be trustworthy [12], [13]. We found calibration ($\text{ECE} \leq 0.004$) more decisive for downstream risk banding than the sub-percent AUC differences between models.

Interpretability as first-class evidence. Agreement between SHAP and permutation importance, and the recovery of textbook epidemiology, give clinicians a reason to *trust* the model. Where a simple, auditable model matches a complex one, the simple model should be preferred [17], [18].

Public datasets carry biases. Self-report, non-response and coverage biases, plus label noise from undiagnosed cases, bound achievable performance and can encode social disparities (e.g. the income effect). Statistical performance and clinical utility are *not* the same quantity [10], [11].

VIII. LIMITATIONS AND THREATS TO VALIDITY

We state the limitations explicitly, as their cumulative effect bounds every claim above.

Observational, cross-sectional design. BRFSS is a one-shot survey, not a cohort. It supports *association*, not *causation*: the SHAP attributions and odds ratios describe how features co-vary with the label under the model, not causal effects. Some associations are plausibly subject to reverse causation—for instance, poor self-rated health (GenHlth) or difficulty walking may partly be *consequences* of diabetes rather than precursors—so the rankings must not be read as actionable causal levers.

Self-report and label noise. Both predictors and outcome are self-reported by telephone, inviting recall and social-desirability bias. Crucially, the outcome is *previously diagnosed* diabetes; the substantial fraction of *undiagnosed* cases is mislabelled negative, attenuating measured associations and likely *under-stating* the attainable signal.

Selection and coverage bias. Telephone sampling under-represents groups without stable phone access and non-respondents, and post-stratification weighting (not used here) only partially corrects this. The observed socioeconomic gradient may therefore encode access disparities rather than biology, with fairness implications for deployment.

Missing variables. No biomarkers (HbA1c, fasting glucose, lipids), no genetic or family-history detail beyond coarse

items, no medication, and only crude diet/activity proxies are available. The plateau we observe is conditional on this impoverished feature space; richer inputs would likely raise it.

Generalisation. Results pertain to U.S. adults surveyed in 2015. Geographic transport (e.g. to other health systems or to the Mexican population) and temporal transport (population drift since 2015) both require external validation and re-calibration before any use [11], [35].

Duplicate rows across the split. Because 9.5% of rows are exact duplicates and the split is random, identical response vectors can appear in both train and test, which may slightly inflate *absolute* discrimination. This affects all models equally and therefore does not change the *comparative* conclusion that is the paper’s core claim.

Methodological caveats. SVM and KNN were tuned on subsamples for tractability, which may modestly understate their performance—though both trail by margins far larger than any plausible tuning effect. Threshold-dependent metrics use the 0.5 cut-off; we mitigate this by ranking on ROC/PR-AUC and by the explicit stratification analysis. We report a single split with bootstrap CIs; complementary multi-seed cross-validated testing [31] is left for future work.

IX. FUTURE WORK

Several directions follow naturally. (1) *Explainable clinical risk stratification*: extend the calibrated banding with per-patient SHAP rationales. (2) *Longitudinal patient modelling*: move from cross-sectional snapshots to trajectories, modelling how individual risk evolves over time. (3) *Temporal risk evolution*: detect improvement/deterioration and early-warning signals from repeated measurements. (4) *Clinical decision-support*: combine risk, evolution and auditable rules into prioritized, justified recommendations. (5) *Integration with digital clinical platforms*: a calibrated, interpretable engine of this kind could be embedded as a decision-support module in a clinical information system such as the PREDIA platform, strictly as a future implementation environment and never as a replacement for clinical judgement.

X. CONCLUSIONS

We benchmarked nine ML estimators for diabetes risk prediction on a large, leakage-free population survey, with bootstrap confidence intervals on a held-out test set. Five conclusions follow. *First*, the available information limits performance more than the algorithm does: nine models of very different capacity saturate at ROC-AUC ≈ 0.83 , a plateau set by coarse self-reported inputs rather than by any estimator. *Second*, complex models offer only marginal improvement—the best model exceeds logistic regression by $\Delta\text{AUC} = 0.007$ (bootstrap CI [0.006, 0.009]), statistically resolvable at this sample size but practically negligible, while leading non-linear models are mutually indistinguishable; reported accuracy (0.866) barely exceeds the 0.861 majority baseline. *Third*, interpretability is clinically valuable: SHAP and permutation importance agree and recover established epidemiology,

favouring transparent models where they match complex ones. *Fourth*, controlling information leakage is essential and dominant: admitting diagnostic biomarkers would swing AUC by ≈ 0.15 , about twenty times the best-versus-simplest gap, so feature-admissibility governance is the decisive decision. *Fifth*, such models support *screening, not diagnosis*: the actionable output is a calibrated probability (isotonic ECE = 0.003) and its mapping to risk bands that separate observed prevalence 16-fold on held-out data, useful for triage but unfit as a diagnostic label and requiring local re-calibration and monitoring. In short, for diabetes prediction from population health indicators, rigour, calibration, interpretability and reproducibility deliver more value than the pursuit of marginal accuracy.

REPRODUCIBILITY STATEMENT

All experiments use a fixed random seed and a single shared, stratified 60/20/20 train/validation/test split; every reported metric is computed on the held-out test set, which was never used for model selection or calibration. The data are the public Diabetes Health Indicators extraction [22] of the CDC BRFSS 2015 survey [23]; no private data were used. The complete pipeline (data audit, training, calibration, risk stratification, explainability and clinical validation), together with every metric table and figure reproduced here, is available as executable notebooks and scripts in the project’s research repository, implemented with scikit-learn [24] and the publicly documented model libraries cited in Section IV.

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